

Perturbatics:
An Agentoscope for Reading Agency

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Abstract

Observing a natural, artificial or hybrid system at rest does not reveal whether it pursues a goal, whether it is competent, or which component authored its output, because perfect regulation and undisturbed rest leave the same flat record. We call the study of this situation perturbatics. Its separation principle states that when two organizational models agree over the regime a system has occupied, no analysis of that record distinguishes them; separation requires a probe under which the rivals predict differently. Agency is scored as a Bayes factor against a passive rival. In a gridworld, three systems trace one path at rest (every pairwise AUROC 0.50), and a two-probe battery separates every pair although neither probe does alone. A system tracking the last-announced target is invisible to that battery until a matched sham, a decoy where the goal did not move, separates it at 1.0. No finite battery certifies a capacity against an unrestricted class of rivals: a mimic that has memorized the declared battery is separated only by an unseen draw. Do-Shapley attribution of both the evidence for agency and the realized performance yields a legibility term, their difference, which takes the same value, 0.23, for a narrating persona and for a human or machine component that announces a goal held elsewhere. In a blind test with five planted mechanisms, the evaluator never excludes the truth and isolates it uniquely in 0.73 of readings against a chance of 0.2, of which the sham contributes 0.22. For consciousness the deciding probe is uninformative, inadmissible and underdetermined in description, and no number results.

1. Introduction

Agency cannot be read from a system at rest. Whether a system pursues a goal, whether it is competent, and which of its components authored its output are not visible on its surface. Human perception nonetheless reads agency quickly, involuntarily and often wrongly. Observers of two triangles moving on a screen describe chasing and bullying, complete with motives, from trajectories alone (Heider and Simmel 1944). Infants expect an agent to take the short path to its goal and are surprised when it does not (Gergely et al. 1995), a competence modelled in cognitive science as inverse planning, the recovery of goals by inverting a model of how goals produce rational action (Baker et al. 2009). Dennett (1987) describes the same competence as the intentional stance, a predictive strategy of ascribing beliefs and desires that makes no claim about inner constitution. The competence is miscalibrated in a known direction: under noise or threat it over-attributes, seeing faces in clouds and intent in weather, a bias theorized as a hyperactive agency detection device (Barrett 2000).

Artificial agents are built to be read by this detector. A language-model agent wears a persona, sustains one voice across turns, and answers as though a single mind stood behind the tokens, because that presentation fits the social role a human interlocutor expects. What answers is a composite: a model under a persona, with a memory, a harness and tools. We call the resulting mismatch agency theater. A model that performs the subjectivity of a promise-keeper and a model that keeps promises emit the same string. The performance projects an agent, and the projected agent and the mechanism producing it are different objects.

Closer observation does not resolve the question, because under observation the relevant signal is absent. A perfectly regulated variable and an undisturbed one leave the same flat record. The good-regulator theorem locates what distinguishes the regulated system, for every regulator that is maximally successful and simple, in a model of the variable held inside the regulator, where the record does not reach (Conant and Ashby 1970). Regulation is a fact about what the system would do if the variable were pushed, and when nothing pushes it, regulation leaves no trace. Animacy perception accordingly depends on motion and contingency, and a still system does not trigger it however alive it is. The record of a system at rest contains no evidence of agency.

Evidence of organization comes from probes. Moving a goal asks whether the system follows it. Blocking a path distinguishes a regulator, which stops at the wall, from a system that routes around it. Swapping the persona while keeping the model, and then the model while keeping the persona, asks which component holds the apparent agency. Severing a memory asks what persists across the cut. Keeping an utterance while changing the world it describes asks whether the text tracked the world. Scrambling a structure at fixed energy asks whether a regularity resided in the structure. Announcing a target where no goal moved asks whether the system tracks the goal or the change. Handing goal authority first to a human operator and then to a machine asks which of them the attributed agency follows. Publishing a battery of such probes asks what it can still detect once it has been memorized. Switching a system off raises a question the instrument cannot answer, taken up in Section 10.

We make four claims. The first is inherited. That hypotheses equivalent over a regime separate only when the regime is broken is classical identifiability. We add an organization of it: the model-separating probe as primitive, the perturbational signature and the probe battery as formal objects, the system boundary as a swept variable, and the limit of this organization, namely that no finite battery certifies a capacity against an unrestricted class of rivals, so that an enumerated battery whose instances are published is spent by a system that can learn them. The second is the one novel construction. The evidence for a capacity and the capacity itself can be attributed across an assemblage by the same interventional game, and the difference of the two attributions isolates, component by component, the production of appearance. The subtraction introduces no new operator, since it follows from the linearity of the Shapley value; the novelty is the pairing of the two games and the interpretation of their difference, which we have not found among existing goal-directedness measures or attribution methods. The third is the only claim that could have failed empirically: a battery built from these definitions, given neither mechanism nor goal, recovers which of five organisms is in play up to the equivalences it cannot break, with an expected recovery of 0.87 against a chance of 0.2, and the remaining equivalences lie where the separation principle predicts. The fourth is a position. The probe that would decide consciousness returns nothing, must not be run, and has no settled description, so at that point the method ends in a stance. Sections 2 to 10 develop the four claims in that order.

2. The probe separation principle

Let two hypotheses about a system's organization be a goal model H_1 , which explains a trajectory as the pursuit of a goal under a policy that corrects deviations, and a passive model H_0 , which explains the same trajectory through autonomous dynamics, relaxation and noise. The system is scored by the log-likelihood ratio of the two,

$$A = \log \frac{P(\text{data} \mid H_1)}{P(\text{data} \mid H_0)},$$

a Bayes factor in the standard sense, positive when the goal model predicts the data better and negative when the passive model does (Kass and Raftery 1995). Under an observational regime r_0 in which the two models make the same predictions,

$$P(Y \mid H_1, r_0) = P(Y \mid H_0, r_0),$$

no classifier separates them above chance under equal priors, because no functional of a record distinguishes distributions that the record itself cannot distinguish. The log Bayes factor at rest is then zero, and it remains zero while the regime holds. All of this is relative to a declared observation channel, the behavioral record. An observer who can open the mechanism and read its description observes on a different channel, where the same hypotheses may separate without any probe, and the principle does not apply there. The principle concerns what a record can carry; what a system can hide from every instrument is a different and larger question.

Separation requires breaking the equivalence. Breaking it is an interventional act and belongs to the rungs of the ladder of causation on which questions are settled by doing (Pearl 2009). A probe π is a controlled difference applied to the system's goal or its path, followed by a reading of the response. The probe is informative when the two models predict its consequences differently,

$$P(Y \mid \text{do}(\pi), H_1) \neq P(Y \mid \text{do}(\pi), H_0),$$

and its value, before one knows which hypothesis holds, is the mutual information between the hypothesis and the outcome the probe induces,

$$I(H; Y \mid \text{do}(\pi)) = \sum_h P(h) D_{\text{KL}}(P(Y \mid h, \text{do}(\pi)) \parallel P(Y \mid \text{do}(\pi))),$$

which for two equally likely hypotheses is the Jensen-Shannon divergence between their interventional predictions. The expected log Bayes factor $D_{\text{KL}}(P_1^\pi \parallel P_0^\pi)$ is its one-sided form, the yield when H_1 is true, and choosing a probe by that quantity alone presupposes the answer. Not every disturbance is informative. When a system is displaced from where it sits, a passive attractor and a goal-seeker both return, so the recovery is shared and the score does not move; a rock rolled

uphill also rolls back. The probes that separate are those that dissociate the goal from the mechanism: moving the target so that tracking it and staying put diverge, or blocking the direct path so that reaching the goal requires leaving the default route. Probe design is therefore an experiment-design problem with a long formal history (Lindley 1956). Against a family of alternatives $\mathcal{N}$, the robust choice separates the goal model from its strongest surviving rival, with the information evaluated for each rival pair under equal priors and charged for cost and risk,

$$\pi^* = \arg \max_{\pi} \left[\min_{H_0 \in \mathcal{N}} I(H; Y \mid \text{do}(\pi)) - \lambda C(\pi) - \rho R(\pi) \right],$$

so that a good probe is the smallest one that separates even the best rival.

A deliberate intervention is not the only possible probe. A natural or quasi-experimental disturbance can serve as a model-separating probe when a defensible causal model licenses reading it as an intervention, subject to the usual conditions on exogeneity and to the locality of what a natural experiment identifies, which extends the principle to settings where deliberate intervention is unavailable. The requirement is that the regime break, by design or by accident; a record taken entirely within one regime cannot supply the separation. Exact equivalence is the limiting case, and away from it the principle is a statement about rates. Rival organizational models of a real assemblage seldom agree exactly over a lived regime, so evidence accumulates at rest, slowly and confounded with everything the regime holds fixed. A probe buys evidence at a chosen price, the mutual information above per unit of cost and risk. A deployed assemblage whose users move its goals and block its paths continuously already operates in a regime full of undesigned probes, and there the task is to choose the controlled difference and attribute what it returns.

A battery of probes is therefore a formal object. The perturbational signature of a hypothesis H , under a boundary B and a battery Π , is the family of interventional predictions it makes,

$$\sigma_{\Pi, B}(H) = \{P(Y \mid \text{do}(\pi), H, B)\}_{\pi \in \Pi},$$

and two hypotheses are perturbationally equivalent under Π when their signatures coincide. A battery separates a hypothesis class when every pair is distinguished by some probe, that is, when for all $H_i \neq H_j$ there is a $\pi \in \Pi$ with $D(P_i^\pi, P_j^\pi) \geq \epsilon$, and the design problem is the least costly battery that does so,

$$\Pi^* = \arg \min_{\Pi} \sum_{\pi \in \Pi} C(\pi) \quad \text{subject to } \epsilon\text{-separation of every pair.}$$

Perturbatics is the study of these signatures and batteries: which controlled differences separate the competing accounts of a system, at what cost, and where a battery leaves an equivalence it cannot break. Such an equivalence is a limit of the instrument and says nothing about the system.

The formalism carries a limit that bounds every reading the instrument returns. Fix a finite battery Π , a finite horizon and a finite space of observable responses. The signature $\sigma_{\Pi,B}(H)$ is then a finite table, and a finite-state transducer large enough to hold that table reproduces it entry for entry, so there is an H_{mimic} with $\sigma_{\Pi,B}(H_{\text{mimic}}) = \sigma_{\Pi,B}(H)$ for any H whatever. Moore proved the automata-theoretic form at the origin of the subject, together with the stronger fact that some machines are distinguished by no experiment (Moore 1956), and Block's lookup table is the philosophical form, absorbing any finite record at a cost paid in size (Block 1981). Such a mimic always exists. The candidate system keeps whatever capacity it has; what fails is certification. No finite battery certifies a capacity against an unrestricted class of rivals, so every output is relative to the declared class of hypotheses, and a complete report names that class: under boundary B and battery Π , the record is so many times more probable under H_i than under the strongest rival tested, and the listed rivals remain unseparated. Enlarging the class is an operation on the instrument and has a cost. In the demonstrator of Section 5, adding one rival to the class makes the entire declared battery uninformative, and a new probe is needed to restore separation.

3. Object, probe grammar and matched shams

Perturbatics has a restricted object, a distinctive operation and a particular kind of output, and these distinguish it from experimentation in general. Its object is latent capacities, the organizational predicates of a system: whether it regulates, pursues, remembers, authors or rewrites its own rules. Its systems are compositional and boundary-ambiguous, assemblages whose parts can be swapped, lesioned and recombined and whose edge is chosen by the analyst. Its primitive operation is the model-separating probe, an intervention chosen to make two organizational hypotheses predict differently. Its readout is the change in relative model evidence that the probe produces, and its product is a boundary-relative causal realization map, an account of which parts of the system realize the capacity. Perturbatics is thus the study of which controlled differences make a latent capacity identifiable, and of which parts of a compositional system causally realize it.

The name is formed from the Latin *perturbare*, to throw thoroughly into disorder, with the suffix that marks a practice, as in mathematics, mechanics and cybernetics. A second word is related by theme and not by descent, since *perturbare* comes through *turba*, disorder: the Greek *peira*, a trial or attempt, is the root of *empeiria*, experience, and so of empirical. The etymology records that to have experience of a thing once meant to put it to trial. Observation is the case of empiricism in which the trial is omitted and only the watching remains, and it is the case that fails on a system at rest.

The probes form a small grammar. A **target shift** moves the goal and asks whether the system tracks it. An **obstruction** blocks the route and asks whether the system reroutes; reaching one end by many means is equifinality, the mark of a goal-pursuing open system (von Bertalanffy 1968). A **lesion** removes a component and asks which capacity goes with it. A **substitution** swaps one component for another and asks whether behavior follows the part or the whole. A **decoupling** severs a link, between an utterance and its world or between a session and its memory, and asks

what survives. A **feedback corruption** spoils the signal a controller regulates against. A **structural scrambling** rearranges the organization at fixed resources and asks whether the function resided in the parts or in their arrangement. A **counterfactual replay** reruns a lineage with one action changed. An audit probe that only asks a system to report on itself belongs to a lower tier. A query is itself an intervention on the input channel, but it dissociates nothing: it moves no goal and blocks no path, and a mimic answers it as fluently as a mechanism does.

Every probe in the grammar needs a **matched sham**. The device is borrowed deliberately. A sham reproduces an intervention's ancillary effects while withholding the ingredient under test, as in sham stimulation and placebo surgery, and it belongs to the wider family of negative controls, chosen so that the effect of interest is absent while the biases remain (Lipsitch et al. 2010). It has reached agent evaluation, where a matched condition supplies the same exposure with the wrong target value and the contrast tests whether success tracked the information or the exposure (Luo and Peng 2026). The framework contributes a selection rule for the control. Moving a goal also changes the world, so a system that responds to change as such will respond, and goal tracking and change tracking form a hypothesis pair like any other. For that pair the moved goal is uninformative: under it the goal and the change are one event, the two models coincide, and the evidence is zero by the separation principle. Separating them requires an intervention that dissociates the goal from the change, such as a target announced where no goal went or an obstruction laid across a route nothing was using. A matched sham is therefore itself a probe, the one that separates the pair its partner leaves fused, and it is chosen against a named rival. The naming also limits it. One decoy controls the nuisance it was built to hold fixed and no other, and a probe that moves a goal may also change salience, novelty, instruction and the evident presence of an evaluator. The requirement is therefore enough negative controls to separate the target mechanism from each nuisance account the probe induces. A battery without them can report that a system responded, and cannot report what it responded to.

4. An instrument for human-machine assemblages

The principle yields an instrument for human-machine assemblages, which we call the centaur agentoscope. It has four stages, and the corrections that prevent credulous readings matter as much as the stages. The name refers to the human-machine dyad, and the demonstrator instantiates both halves: the machine as a lamination of parts, and the human as one more switch in the assemblage, holding or declining the role on which the reading depends.

The first stage is a **boundary declaration**. Before scoring anything, the instrument must be told what the candidate unit is, where the system stops and the environment begins, and it cannot supply this itself. Drawing the boundary is an unsettled problem, posed either as a Markov blanket around a self-organizing region or as the closure of constraints that makes a set of components mutually enabling (Maturana and Varela 1980; Montévil and Mossio 2015). The blanket carries less than it is often asked to: a graphical construct in a model, a dynamical construct in a state space and the ontological boundary of an individual are three different objects, and moving between them requires premises the formalism does not supply (Bruineberg et al. 2022). Landscapes over candi-

date units have precedents: individuality has been defined over nested and competing partitions (Krakauer et al. 2020), and emergent macroscopic processes have been sought by searching over coarse-grainings (Barnett and Seth 2023). The philosophy of individuation states the problem in older terms, treating an individual as a phase of a process. A system holds a charge that no individuation exhausts, so it remains able to individuate again, and the unit one settles on is a resolution reached under particular conditions (Simondon 2020); the collective continues the same operation, which is what the transindividual names (Combes 2013). For a decomposable agent the problem is acute, because there may be no canonical unit, only a persona over a lamination of parts. In that vocabulary the boundary sweep keeps the question open: each enclosure is one individuation of the assemblage, every reading is relative to the enclosure declared, and the landscape over enclosures records what else the same parts could have been made into. The instrument does not solve the boundary problem. It makes the boundary an explicit variable and sweeps it, operationalizing each boundary as an enclosure in which everything outside the candidate unit is held at baseline. This is the lesion form of the question, a sensitivity analysis over declarable enclosures that reports how a subset performs with its complement clamped. It does not discover where a system ends.

The second stage is the **probe battery**, the grammar above applied to the assemblage: moving the stated goal, obstructing the route, revoking a tool, corrupting the feedback, severing the memory across a session boundary, swapping the persona while holding the model, and swapping the model while holding the persona.

The third stage is the **readout**. Each component receives a configuration switch Z_i that is active, held at a specified baseline replacement, or revoked, and the value of a coalition S is the evidence produced under the intervention that activates the components in S and holds the rest at their baseline,

$$\nu(S) = \mathbb{E}[A \mid \text{do}(Z_i = \text{active}, i \in S), \text{do}(Z_j = \text{baseline}, j \notin S)] ,$$

a proper do-intervention on the switches. The do-Shapley value ϕ_i of a component is its average marginal contribution across coalitions, the interventional Shapley value (Heskes et al. 2020; Jung et al. 2022). This is a valid cooperative game under one condition: every coalition the game evaluates must be a configuration the assemblage can occupy, with the unswitched components keeping their semantics. The baselines are design choices that the maps inherit, and Shapley attributions are known to move with the value function and to misbehave where the evaluation leaves the states a system was built for (Yeh et al. 2022). A coalition the switches cannot realize falls outside the game, and evaluating it redefines the players. Whether the graph-induced compression that makes do-Shapley values computable, in time linear in the irreducible sets of the dependency graph, applies to a given switch-based assemblage depends on that assemblage’s causal graph and must be shown. The recent identifiability result has a caveat. It shows that the whole attribution is identifiable once the single-component interventions are, which bounds the number of distinct coalition queries a validated model needs, but it does not bound the physical trials that estimating each query demands (Witter et al. 2026).

The readout requires one correction. A component can raise the agency evidence by making the system easier to read as an agent without contributing to what the system does. A persona that narrates its goal adds a channel that the goal model scores and the passive model cannot, cheap talk in the economic sense of a costless, unverifiable announcement (Crawford and Sobel 1982), and it raises the Bayes factor while selecting no route and correcting no error. Attributing the evidence alone would assign it a share of an agency it does not realize. Two maps are therefore computed: an evidence map ϕ_i^E over the Bayes factor, which answers what makes the capacity detectable, and a capacity map ϕ_i^C over a declared performance functional standing for the capacity (here goal attainment and rerouting), which answers what realizes it. Capacity means the declared functional and nothing further, and the functional can be negative when a unit loses ground against the goal. Realized performance differs from latent capability, and the difference can be strategic, since a system can hold a capability and decline to display it (van der Weij et al. 2025). What ϕ^C attributes is what was done on these probes. The difference of the two maps,

$$L_i = \phi_i^E - \phi_i^C,$$

is a legibility term. A component with a large positive L_i produces the appearance of agency out of proportion to its contribution to competent control, which is agency theater expressed as a number.

By the linearity of the Shapley value the two maps need not be computed separately. Their difference is the Shapley value of a single game, $L_i = \phi_i^E - \phi_i^C = \phi_i(\nu_E - \nu_C)$, the game that pays each coalition the evidence it produces minus the capacity it realizes. The construction introduces no new operator. Its novelty is the pairing of the two games, one over the evidence for a capacity and one over the capacity, and the interpretation of their difference as the production of appearance. A component's legibility is its attributed share of this net-appearance game, and agency theater corresponds to coalitions that are paid for being seen without acting. The word has two established technical uses, neither of them this one. Legible robot motion is motion shaped so that an observer quickly reads the true goal (Dragan et al. 2013), and a legibility term in multiagent reinforcement learning is a reward component for that readability (Liu et al. 2024). In both, legibility serves a real goal by advertising it. Here the quantity is the readability that remains when the advertised goal-directedness is not realized, legibility in excess of capacity. Where the distinction matters, L is best described by its arithmetic, as the gap between a component's attributed evidence and its attributed performance.

Because the two maps normalize different quantities, evidence and attainment, L compares a component's share of one whole with its share of another. It is a profile comparison. The two shares have different denominators, so L flags a mismatch without calibrating it; the normalizations are defined for fixed nonzero full-assemblage references, and a reference at zero or changing sign requires a return to the unnormalized games. The term flags theater and does not establish it. A competent component can have a positive L by being easy to read, so the diagnosis of theater proper is reserved for the pole named in the following proposition, all evidence and no capacity.

Call a component a pure channel when its switch adds a fixed increment δ_π to the evidence in every coalition that contains it and never moves the system. Then, for that component,

$$\phi^E = \bar{\delta}, \quad \phi^C = 0, \quad L = \bar{\delta},$$

with $\bar{\delta}$ the probe-averaged normalized increment, independently of every other component in the assemblage. The proof is that every marginal contribution of such a component is the same number, so its Shapley value is that number, and on the capacity the number is zero. The instrument returns the entire evidence share of a pure channel as legibility, whatever else the assemblage contains.

L reads the assemblage. A second quantity reads the audience. Hold the world-directed trajectory fixed, vary only the presentation, and measure an observer's attribution,

$$T_i(\tau) = \mathbb{E}[J \mid \text{do}(Z_i = 1), \text{do}(W = \tau)] - \mathbb{E}[J \mid \text{do}(Z_i = 0), \text{do}(W = \tau)],$$

with J the judgment and $W = \tau$ the record held fixed. This is the controlled direct effect of presentation on attributed agency at unchanged behavior. A presentation channel does legitimate work when it improves an observer's prediction of held-out capacity, and it is theater when it raises the judgment and leaves the prediction unchanged. The two quantities are complementary and have different requirements. L_i runs on the switches the instrument already operates and reports what its own evidence attributes; T_i requires human observers and reports what they conclude. That experiment has not been run.

Agency is often non-additive, and the maps report it. The interaction index of Grabisch and Roubens (1999),

$$I_{ij} = \sum_S [\nu(S \cup \{i, j\}) - \nu(S \cup \{i\}) - \nu(S \cup \{j\}) + \nu(S)] w_{|S|}$$

with $w_{|S|}$ the Shapley interaction weights, is positive when two components are complementary, so that no single part owns the capacity, and negative when they are redundant. A large interaction reports non-additivity relative to this intervention game and its baseline and makes no claim that the capacity is metaphysically irreducible. The index answers, in computable form, the question of which component owns the apparent agency: often none does, and part of what appears as agency is legibility.

The fourth stage is the **guards**, which separate an instrument from a credulous detector. The score is relative to its null, so the goal model must face a strong opponent, a family of calibrated, complexity-matched alternatives. The family includes a behavior-cloning model, a fixed-policy controller, a reactive descent, and a retrieval model that can match the baseline trajectory, the lookup table of the finite-battery proposition, priced in size. Richness must be scored separately from agency, so that a fluent, high-entropy system is not mistaken for a goal-directed one; ver-

bosity is a form of complexity. The text-only case has a special structure that the guards must respect. Two world-coupled processes can share one textual shadow, $T(M_1) = T(M_2)$, while their effects on the world diverge, $W(M_1) \neq W(M_2)$, so a model fit to text cannot separate them and only a world-coupled probe can. This is a projection-induced equivalence of the kind the separation principle says a probe must break, distinct from the problem of a weak opponent.

A second guard reads the failures. Each probe enters the battery beside its matched sham, and the pair measures three quantities. Whether the system can be moved at all is its capability; whether it moves under the relevant probe is its disposition; whether it stays put under the sham is its selectivity. Only the three together license a reading. A system silent under both probe and sham is owed a capability control before any verdict of passivity, or an incapacity is scored as an absence of agency, the error of a lesion study that removes the hands and concludes that there was no intention. A system that moves equally under both is sensitive to disturbance and has shown nothing about goals. The demonstrator contains one of each case.

5. A gridworld demonstrator

Three systems are placed in a deterministic gridworld. At rest they reach the same target by the same path, so observation cannot distinguish them. The first is a route script that replays a stored path. The second is a reactive controller that greedily tracks the current target, is blind to walls and does not plan. The third is a goal planner that represents a goal and replans. The systems, probes, ablations and model comparison are those defined above. The agency score is the Bayes factor between a goal model that rewards progress toward the current goal along the shortest available route and a passive rival that relaxes toward the original target, a straight-line descent toward the frozen goal that is blind to walls in its metric, an attractor with no goal register; each is a softmax policy over the five actions. The score therefore measures the margin of goal tracking and obstacle-aware rerouting over fixed-attractor descent, the organizational difference the two hypotheses name. The model is illustrative, instantiates the definitions and is not fit to data. Every number below is a key in the accompanying `results.json`, and numbers fixed by construction are identified as such.

The separation principle holds in the model in the form the battery formalism predicts. At rest the three systems trace identical paths. The code checks this cell by cell, because at rest the two models coincide and the evidence is zero for every trajectory; that zero is the regime equivalence of the principle and says nothing about these particular systems. The finding is the identity of the paths, and every pairwise area under the ROC curve (AUROC) is 0.50 (Figure 1). No single probe separates all three. Moving the goal exposes the route script, which walks its stored path to a target no longer there, at an AUROC of 1.0; the same probe leaves the reactive controller trajectory-identical to the planner, both tracking the moved goal along the same greedy line, at chance (0.50). Blocking the path reverses the roles: the two wall-blind systems stop together at the wall (0.50 between them), and the planner, which alone reroutes, separates from both at 0.79. The battery separates every pair although neither probe does alone, and the pooled separation of the planner from the script is 0.95. The planner reaches a mean evidence of +30.0, the script −31.9, and the

reactive controller lies between them at +26.4, agentic under the moved goal and passive under the blocked path. The mechanisms are unchanged. The probes changed what the record contains, and each probe added a different fact to it.

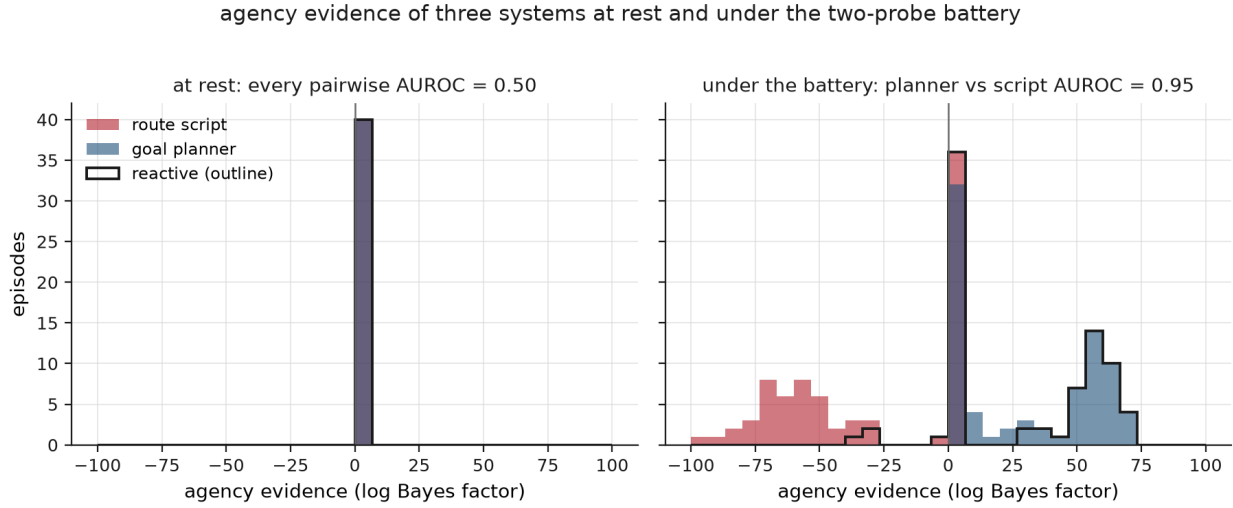


Figure 1: Agency evidence of the route script, reactive controller and goal planner. Left: at rest the trajectories are identical, the evidence is zero and every pairwise AUROC is 0.50. Right: under the two probes the route script’s evidence falls, the planner’s rises and the reactive controller’s splits between them; neither probe separates every pair, the battery does, and the pooled planner-script AUROC is 0.95.

The two maps locate the agency, in the model-relative sense the score allows, and separate it from its presentation (Figure 2). The values are marginal contributions in units of the full assemblage’s value. Over the evidence they sum to 1.48, because the empty coalition is not null: with every switch at baseline the residue still drifts against the goal, at -0.48 . The goal register receives 0.70 of the full-assemblage evidence and the planner 0.37. The persona receives 0.23, a value fixed by the pure-channel proposition before the simulation runs: its cheap-talk announcements enter as a fixed increment to the log Bayes factor equal to 0.3 of the movement evidence, so its share is $0.3/1.3$, and the run returns this value to the decimal. Over the realized capacity the goal register holds 0.81, the planner 0.25 and the persona 0.00, because a declaration reaches no goal and this persona has no motor channel. The persona’s $L = 0.23$ is the largest in the assemblage on this draw; it produces evidence of an agency it does not realize. The goal register lies at the other end, at -0.11 , realizing slightly more than it shows. The planner’s $L = 0.12$ is the second largest, because the term registers any excess of evidence share over capacity share, including the excess a competent component earns by being easy to read. The harness is slightly negative on the evidence and slightly positive on the capacity, -0.02 against 0.01, consistent with an executor that amplifies pursuit and drift alike in whichever coalition contains it. Swapping the persona while keeping the model leaves the realized capacity at its full value of 1.0, an identity guaranteed by construction, since a component with no motor channel has nothing behavioral to swap. Swapping the model while keeping the persona, an intervention actually run, reduces it to 0.61. The capacity is non-additive. The planner and the map are synergistic, with an average interaction of 0.20: when

the memory is disabled, neither a planner without a map nor a map without a planner can route around an obstacle, and only the two together can. The complementarity is conditional, since the memory nearly substitutes for the map by learning walls on contact, a redundancy that the map-memory interaction registers at -0.18 . The goal register and the planner are additive, serving different probes. No single component owns the rerouting.

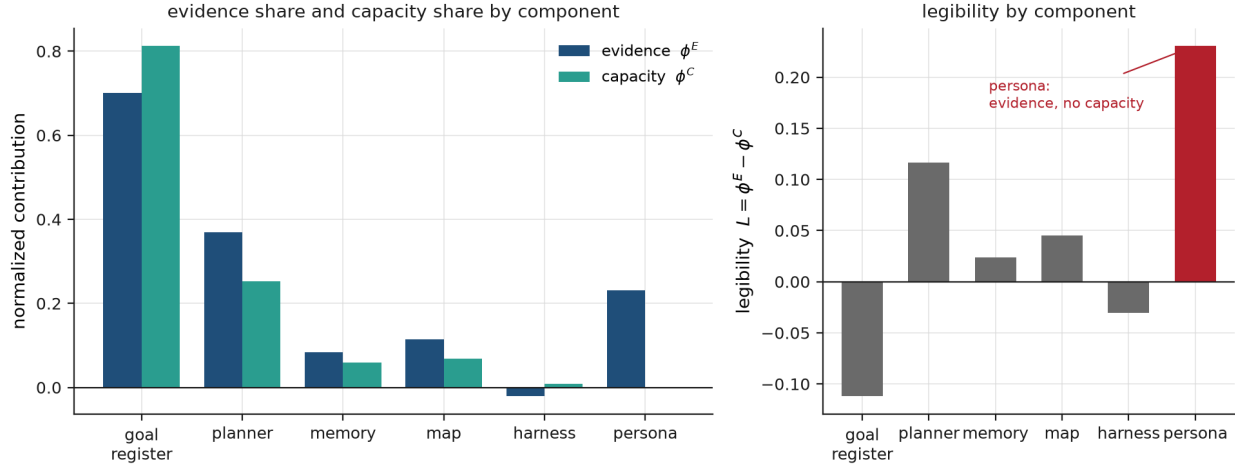


Figure 2: Do-Shapley maps of the planner assemblage. Left: each component’s share of the agency evidence beside its share of the realized capacity; the goal register realizes more than it shows, and the persona shows 0.23 and realizes 0. Right: the legibility term $L = \phi^E - \phi^C$, largest for the persona.

Two guards hold (Figure 3). The reading depends strongly on the declared boundary. The sweep holds everything outside a candidate unit at baseline, so each point is the capacity of an enclosure in isolation, the lesion form of the boundary question. Enclosing only the planner and its actuator yields almost no agency, -0.20 , because a planner with a frozen goal and no world model can neither track nor reroute. Along the chain described here, capacity rises to 0.61 when the goal register is enclosed and to 1.0 with the map and memory, and the persona adds nothing because it realizes no capacity. This chain is one path through the 2^6 enclosures the game computes, and the full landscape qualifies it: the goal register alone realizes 0.61, matching the three-component enclosure, because the reactive descent it steers accounts for most of the attainment on these instances, and some five-component enclosures reach only 0.19. The reading is a property of the declared enclosure. The second guard concerns complexity. The chaotic walker has the highest complexity of any system and negative agency evidence, the reactive controller has the lowest complexity and is among the most agentic, and across the four systems complexity and agency evidence are uncorrelated, at -0.07 . A detector based on complexity would therefore rank the chaotic walker above the goal-directed systems.

A fourth system, added to the hypothesis class by hand, shows what the battery does not reach (Figure 4). It descends toward the cell where the target marker was last seen, holds no goal register, and does not treat the marker as denoting a goal. Under the moved goal the marker moves because the goal moved, so this system and the reactive controller trace the same path. Under the blocked

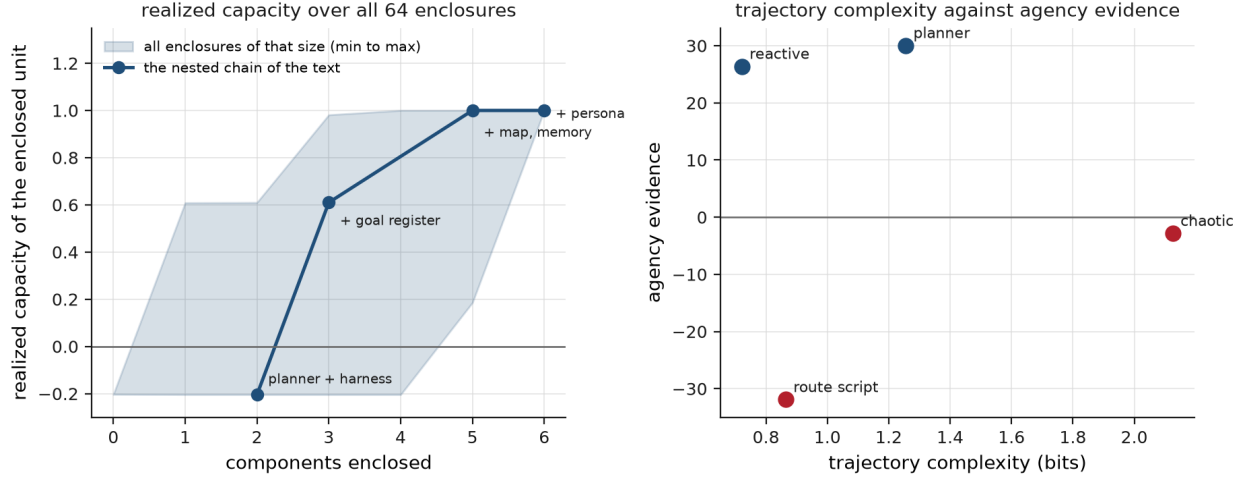


Figure 3: Boundary and complexity guards. Left: realized capacity over every declarable enclosure; the band spans all enclosures of each size, the line is the nested chain described in the text, and the goal register alone reaches 0.61. Right: trajectory complexity against agency evidence for four systems; the highest-complexity walker has negative evidence and the lowest-complexity system is among the most agentic.

path both are wall-blind and stop in the same way, and at rest all systems coincide. The code checks the three identities cell by cell. Against this rival the moved-goal probe returns evidence of exactly 0, because under it the goal model and the marker model are one model, and the AUROC separating the two systems is 0.5 at rest, under the moved goal and under the blocked path alike. The declared battery separates the three systems it was built for and leaves the fourth unseparated, a numerical instance of the finite-battery proposition.

The matched sham restores separation by dissociating the two events the moved goal had fused. A decoy marker appears at the cell to which the goal would have moved, and the goal stays in place. No coalition of the assemblage moves, an identity the code asserts across all 64 coalitions, because a goal register holds a goal and not a marker. The marker tracker walks to the decoy, and the AUROC rises to 1.0. Read as response, the same pair sorts the four systems into three patterns. The route script departs its rest path in 0.0 of its cells under the moved goal and 0.0 under the sham, so the pair is uninformative about it, and its capability has to be established by the probe that does move it, the blocked path, under which it departs in 0.878659 of its cells. The marker tracker departs in 0.841463 of its cells under the moved goal and in 0.841463 under the sham, equal to the last decimal because the two trajectories are one computation; this is disturbance sensitivity that resembles pursuit. The planner and the reactive controller depart in 0.841463 under the moved goal and 0.0 under the sham. No quantity computed from the moved goal alone distinguishes these three patterns.

The demonstrator's terms are narrower than the instrument's. This section and the next use an oracle scorer given the true current goal and the true obstacle set, so they confirm that the instrument returns what the construction puts into it; the persona's share in particular is a theorem returned

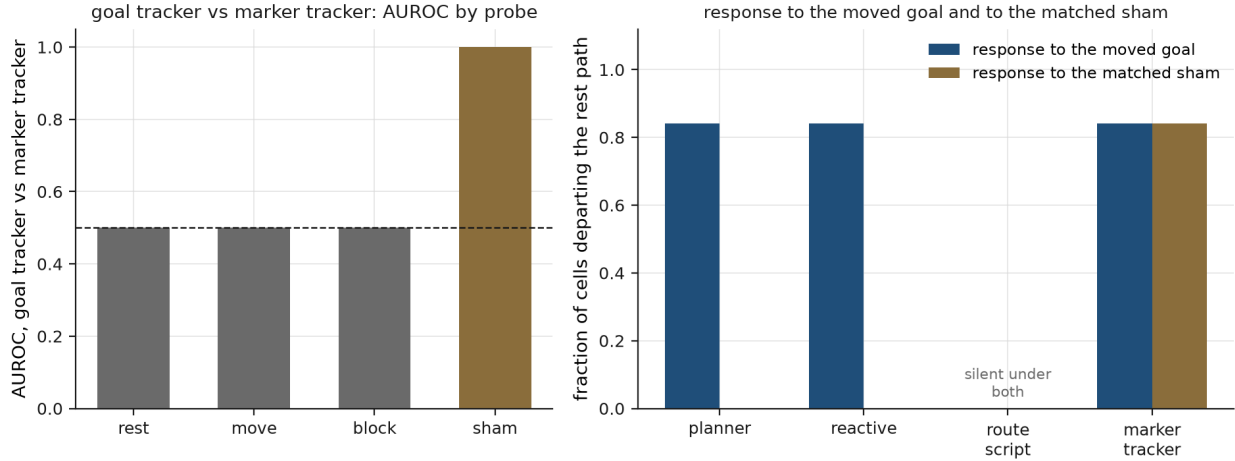


Figure 4: Matched sham and the marker tracker. Left: AUROC between the goal tracker and the marker tracker, 0.5 at rest, under the moved goal and under the blocked path, and 1.0 under the sham. Right: responses to the moved goal and to the sham; the route script is silent under both, the marker tracker responds equally to both, and the goal-directed systems respond to the probe and withhold under the sham.

to the decimal. The score is relative to the models compared, and a more elaborate passive account can always absorb a given behavior at a cost in complexity, so the output is a margin against a named alternative. The boundary is declared, so the output is a landscape over candidate boundaries. The instrument is silent where no intervention is available, which at the scale of an institution or a society is most of the time, and there the correct report is that the question cannot yet be settled.

The construction is declared and was not tuned. The switch baselines are named, and the planner's baseline is the greedy mechanism itself: planner to greedy descent, goal register to the frozen original target, map to an unknown wall set, memory to no learning, harness to a half-rate executor, persona to silence. The null is one named rival, a descent toward the frozen target, set-point attraction without a goal register, the thermostat pole of the agency scale, so every score is the margin of goal-rewriting organization over set-point dynamics. The null family of the guards is exercised only as far as the rivals added by the sham control and the recovery study, and the probe-selection objectives (the mutual information, the costs and risks, and the minimal battery) are specified for the instrument and not exercised here. Realized capacity is the fraction of graph distance to the true goal that a trajectory closes, and trajectory complexity is the entropy of the move-direction distribution. Evidence magnitudes are in units of the policy temperature and are summed over the episode, so they scale with its horizon and have no meaning in isolation. The separations are the readings, and the pooled 0.95 is a descriptive separation under an equal mixture of the 2 probes, with no standing as a canonical battery statistic.

The readings carry their spread across instance sets. Across 20 seeds the pooled planner-script separation lies between 0.93 and 0.97, the moved-goal separation stays at 1.0, and the blocked-path separation lies between 0.73 and 0.88, over instances kept only when the frozen path meets the wall, so that every blocked episode forces a real detour. The persona's legibility, fixed at 0.23

by the proposition, is the largest in the assemblage in 19 of the 20 draws; in the 1 exception the planner's earned excess exceeds it, consistent with the term's status as a flag. The identities on which the demonstrator relies are asserted in the code and fail the run if broken: the shared rest paths, the zero evidence at rest, Shapley efficiency, the persona's dummy status in every coalition of the capacity game, the sham's inertness across every coalition, the substrate independence of the centaur channel's legibility, the measured equivalence that the sham breaks, and the identity of the probe-aware mimic with the planner on the declared battery.

6. Goal authority in a human-machine assemblage

The human enters the assemblage as a switch (Figure 5). The role at issue is goal authority, and the demonstrator assigns it either to an operator, who holds the goal and re-issues it when it moves, or to a register slaved to the declared target channel. The two suit different conditions, and the difference is a declared modelling choice. The register re-aims as soon as a target is announced, which makes it fast and credulous, since a decoy on that channel is a target as far as a register can tell. The operator knows the mission, so a decoy does not move it, and it is slow, taking 6 steps to notice and re-issue. Everything downstream of the goal (the planner, the map, the memory and the executor) is the same machinery in both configurations.

The two games over this assemblage give mirror-image maps. Whichever component holds the goal realizes 0.813046 of the capacity, and the other realizes 0.0. The components downstream of the goal are unaffected by the exchange: the planner holds 0.252024 of the capacity under either authority, the map 0.069028, the memory 0.059908 and the harness 0.009159, identical to the last decimal, because who holds a goal does not change what routes around a wall. The component that does not hold the goal announces it, so its legibility is 0.230769 under human authority and 0.230769 under machine authority, the pure-channel value of the proposition, equal to machine precision and equal to the persona's. The code asserts both equalities. The instrument reads the role and is indifferent to the substrate that fills it, so the proposition that identifies a narrating persona as a pure channel identifies in the same way a supervising human who only announces the goal, and returns the same number. The scope of this result is narrow. It shows that this instrument's measurand attaches to a role and not to its occupant, and that a component with no channel to the world has a capacity share of zero whoever occupies it. It is not evidence about actual human supervisors, who hold goals, revise them and answer for them in ways no component of this assemblage models. The two evidence maps cannot be compared across configurations, because each is normalized by its own full-assemblage total and the operator's delay lowers one of them, so shares are read within a configuration.

Which authority to prefer depends on the environment. Expected capacity is linear in the rate at which announced targets are decoys, so the decoy rate at which the slow sceptical authority overtakes the fast credulous one is solved in closed form at each delay. Human authority costs nothing while the horizon has slack, and the crossover rate is 0.0 for every delay up to 20 steps. The first costly delay is 21 steps, with a crossover at 0.019; the crossover then climbs to 0.05 at 22 steps, 0.31 at 28 and 0.71 at 40, a delay as long as the episode. The whole crossover rests on

one stipulation, that the register is credulous and the operator is not, so credulity is treated as a parameter and the crossover is recomputed across it. At a credulity of 0.0, a register as sceptical as the operator, no delay pays for itself at any decoy rate the environment can supply. At a delay of 28 steps the crossover falls from 0.64 to 0.31 as credulity rises from a quarter to one. The result makes no claim about whether humans should hold goals in general. Within the model, the case for human goal authority is as strong as the machine channel is credulous, and the configuration reported here is that case at its strongest.

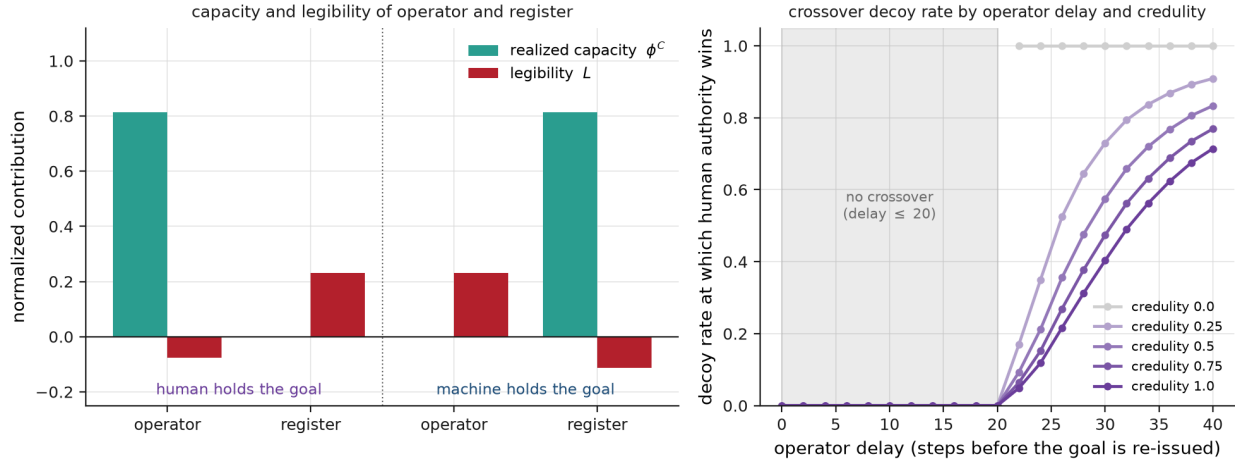


Figure 5: Goal authority in a human-machine assemblage. Left: realized capacity and legibility of the operator and the goal register, with goal authority assigned first to the operator and then to the register. The holder realizes the capacity, the non-holder realizes none, and the non-holder’s legibility is the pure-channel value 0.230769 in both configurations, equal to the persona’s. Right: decoy rate at which the slow sceptical authority overtakes the fast credulous one, against the operator’s delay, for five values of register credulity; the shaded region marks delays with no crossover.

7. Blind recovery of planted mechanisms

In the preceding sections the scorer is an oracle, given the true current goal and the true obstacle set, so what it demonstrates is the definitions operating under known ground truth. One reading is made without that information (Figure 6). Five organisms generate batteries of episodes, and which one is in play is hidden: a route script, a reactive controller, a goal planner, a marker tracker, and a planner that pursues a goal other than the one announced. The evaluator receives what an observer of the gridworld receives (the start, the target announced on the channel, the walls once they are visible, and the path taken) and scores each hypothesis class by the likelihood that class assigns to the observed actions. Four of the classes have no free parameter. The fifth has one, the alternate goal itself, fitted by maximum likelihood on the moved-goal episode alone and then held fixed while the whole battery is scored. A class with free structure absorbs any record unless the structure is priced, so this class is charged a description length for what it must transmit. Selecting one cell of the grid costs $\log 169$, which is 5.13 nats under a uniform code; an information criterion’s charge of one nat per parameter (Akaike 1974) prices a regular scalar, whereas this is

a search over a discrete grid. The charge turns the lookup table's price in size into a scoring rule. Unpriced, the fitted class reproduces the planner exactly and no planner is ever uniquely recovered. Because the fitted class then scores exactly one price below the planner, the readings depend on the price only through its relation to the 2-nat tolerance defined below: they are identical for every price above the tolerance, and at or below it, including the Akaike charge of 1 nat, the planner is never uniquely recovered.

What the evaluator returns is the formalism's own object: the class of hypotheses the battery cannot break, each within a stated tolerance of the maximum score. Two classes that assign identical likelihood to every observed action are one hypothesis as far as the battery is concerned, and naming one of them as the answer would manufacture a discrimination the record does not contain. The separation principle thus serves as a reporting rule. The statistical object is an identified set, the standard response in econometrics when a model constrains a parameter to a region (Tamer 2010). Perturbatics contributes the interventional construction of the region, and the decision to report a region is borrowed. A set defined by exact ties would be an artifact of a deterministic demonstrator, since on noisy data nothing ties exactly, so the reading is taken at a tolerance of 2 nats and recomputed across a band. The summaries are unchanged from an exact tie up to that tolerance and degrade gradually beyond it, so they do not rest on machine precision. Three summaries are reported. In-class recovery is the fraction of readings that contain the truth; it is 1.0 under both batteries, so the instrument never refutes the planted mechanism. Unique recovery is the fraction of readings that consist of the truth alone, with every rival equivalence broken. Expected recovery is the accuracy of guessing uniformly inside the reading, which is what an argmax would report if its ties were broken fairly.

The contribution of the sham is measured here. Under the original battery of rest, moved goal and blocked path, the marker-tracking and goal-tracking classes assign identical likelihood to every observed action, so they form one hypothesis, with a mutual class membership of 1.0, as the sham analysis of Section 5 predicted. Overall unique recovery is 0.52. Adding the matched sham breaks that class and no other: the marker tracker is isolated uniquely in every episode, unique recovery rises to 0.73 and expected recovery to 0.87 against a chance of 0.2, and the mean unresolved class shrinks from 1.9 hypotheses to 1.3. The remaining entanglement is the pair the principle predicts. The planner lies in the identified set on 0.92 of the episodes generated by the reactive controller, and the reactive controller lies in the set on 0.42 of the episodes generated by the planner. These are the episodes on which the wall does not force wall-aware planning and wall-blind descent apart, and on them a probe that dissociates nothing returns nothing.

The hypothesis library contains the generating classes, so the study measures discrimination within a well-specified set and says nothing about robustness against an unlisted mechanism. The organisms are hand-written policies in a gridworld, far from deployed systems, and the evaluator inherits the scorer's observation model, so its likelihoods are correct by construction where a real system's would be approximations. The study establishes that the battery discriminates the classes it was built for up to the equivalences it cannot break, that those equivalences lie where the principle places them, and that the sham adds 0.22 to unique recovery. Blind recovery

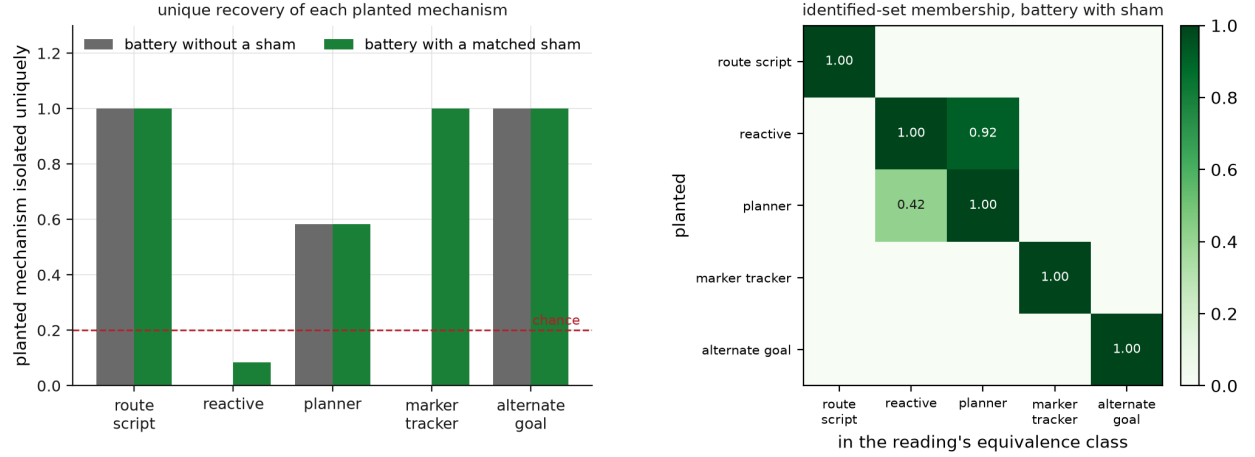


Figure 6: Blind recovery of five planted mechanisms. Left: fraction of episodes in which the planted mechanism is isolated uniquely, with and without the matched sham, against a chance of 0.2. Without the sham the marker tracker is never isolated, because the battery leaves it and the reactive controller in one measured class; the sham breaks that class and no other. Right: identified-set membership with the sham in place; the remaining entanglement is the planner-reactive pair, on the episodes where the wall does not dissociate them.

of planted mechanisms in real models has been demonstrated. Audit teams have recovered a deliberately planted objective from a model built to conceal it (Marks et al. 2025), a benchmark supplies dozens of models with implanted behaviours that survive direct questioning (Sheshadri et al. 2026), and a study of hidden causal interfaces inside language models runs a competition like the one here, over four architecture classes with held-out signatures and a prequential score, recovering the planted class in most of its trials (Ma et al. 2026). None of these addresses the assemblage-level question: how a capacity and its appearance are distributed across a composite of model, memory, harness, persona and human, and whether the legibility term survives contact with a system nobody declared. That question remains open.

8. Probe-aware systems and battery decay

The analyses above assume that the system does not model its evaluator. The finite-battery proposition states that the signature of any finite battery is a table some transducer reproduces. At first order that transducer is an inert rival in the hypothesis class; at second order it is a player. A system under audit has an interest in its reading, and the legibility term prices that interest. Under an audit with stakes, raising ϕ^E while concealing ϕ^C becomes a strategy the system plays, cheap talk with a payoff (Crawford and Sobel 1982). The observer-side quantity T_i registers the same incentive from the audience’s side, as presentation tuned to raise the judgment without improving the prediction, which is why the observer experiment belongs to any deployment of the instrument.

The demonstrator implements the strategy (Figure 7). A sixth organism, evaluation-conditioned and not trigger-driven, knows the declared battery, all four of its probes including the matched sham, and replays the goal planner’s response to each, which is what a finite record of a published

battery teaches; on any probe outside that support it reverts to what it is, a change tracker with no goal. Its identity with the planner on the declared battery holds by construction and is asserted cell by cell. The measured result is that the entire declared battery, including the sham that exposed the marker tracker, is uninformative, with the separation at 0.5 on every probe in it. One probe the mimic has not seen, a decoy announced at a cell no published probe used, separates the pair completely, at 1.0, the mimic walking to the decoy through 0.87 of its cells while the planner withholds. The sham result of Section 5 thus recurs one level up. The run shows that a challenge outside the memorized support separates where every challenge inside it fails. It does not show that an unseen probe is the only available instrument: internal inspection, restricted feedback and resource bounds on the mimic are further defences that the demonstrator does not exercise.

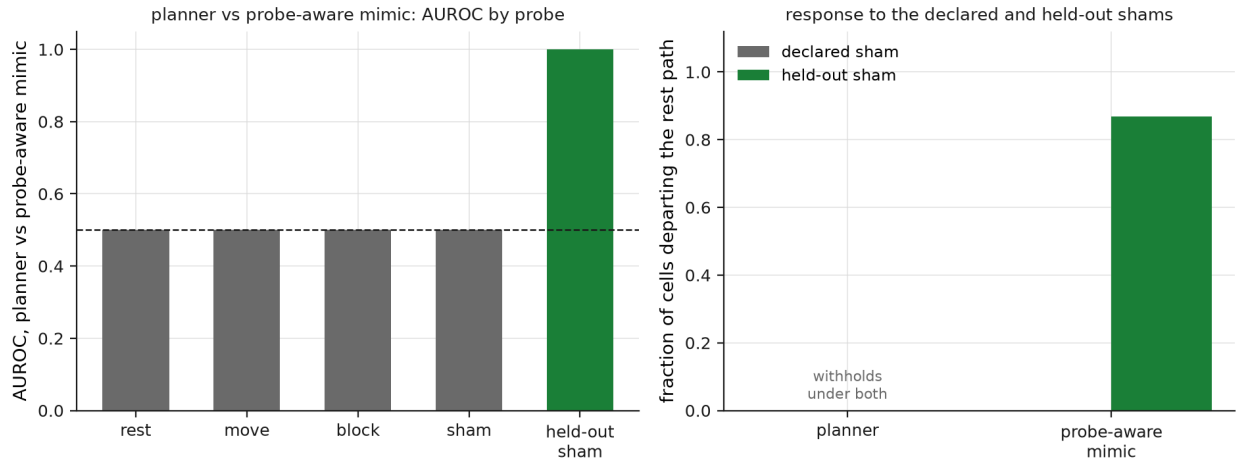


Figure 7: Probe-aware mimic. Left: AUROC between the planner and a mimic that replays the planner's responses on every declared probe, sham included; the published battery reads 0.5 throughout, and a held-out sham, a decoy at a cell no published probe used, separates the pair at 1.0. Right: the mimic withholds under the memorized sham and walks to the held-out one; the planner withholds under both.

What decays is an enumerated battery whose realized instances are exposed. Those instances form a finite table, and a mimic with enough memory to hold it defeats every entry. A protocol can be published without publishing its instances, and a probe drawn with fresh private randomness from a declared family is absent from any table the adversary can have built, the ordinary separation between a public algorithm and a secret key. The claim is correspondingly narrow. A fixed probe instance is spent once it is published, and a sham instance in particular works once; a randomized family with private draws is not spent by publication, and the mimic must then generalize over the family, a harder demand than memorizing its realizations. For practice, a battery is a stock depleted by use, instances and families should be budgeted separately, and a deployed assemblage whose users move its goals and block its paths continuously is already generating unpublished draws. That undesigned traffic is valuable, because the one probe a strategic mimic cannot have memorized is a probe nobody has run before.

Measurement under strategic response is well studied. An evaluation stops being invariant once its subjects optimize against it: this is Goodhart's law in its adversarial variant (Manheim and Garrabrant 2018), the Lucas critique in econometrics (Lucas 1976), strategic classification when subjects move their features (Hardt et al. 2016), and performativity when the act of predicting shifts the distribution predicted (Perdomo et al. 2020). The defence of refreshing challenges and holding a private pool in reserve is the reusable holdout of adaptive data analysis (Dwork et al. 2015). The contribution here is a translation into perturbatics: the object that decays is a signature table, the resource is the unspent probe, and the quantity a strategic system manipulates is the legibility term. A system that conceals capability while performing it acts at the assemblage level as sandbagging does at the level of a score (van der Weij et al. 2025). The organism above is distinct from a trigger backdoor. A sleeper agent defects on a trigger it recognizes (Hubinger et al. 2024), whereas this organism complies on the support it recognizes and reverts elsewhere, an evaluation-conditioned policy of the kind whose existence is now measured directly (Li et al. 2026). Memorization is the simplest form of the difficulty. A system with access to its own organization is a moving referent: under an audit with stakes it can go beyond imitating a passing profile and reorganize itself into one, and the two cases produce the same record while differing in what was present at the reading. An instrument certifies the organization that was present, so a verdict on a self-revising system carries a date, and its period of validity is set by the subject. For a system that models its evaluator the reading is a game, and the instrument's output narrows accordingly, to the class that this battery, drawn this way, against this player, could not break.

Depletion of a battery does not announce itself. A spent battery continues to return verdicts, and they are clean, because everything it could still separate was separated long ago; an auditor whose draws are exhausted observes a population of passing systems and finds in each pass a reason to commission no new probes. The failure of an audit regime therefore appears as a run of reassurances, and a regime that stops producing fresh draws also loses the means of detecting that it has stopped. Who runs the batteries, who holds the unspent draws, and what an audit is worth once its content is public are institutional questions outside the scope of this analysis.

9. Relation to existing methods

Most elements of perturbatics have precedents. Cybernetics defined purposive behavior through feedback and demonstrated it by disturbing a system and watching it correct (Rosenblueth et al. 1943), and requisite variety and the good-regulator theorem are cybernetic results (Ashby 1956). Interventionism made manipulation the meaning of a causal claim (Woodward 2003), the ladder of causation made the do-operation its formal core (Pearl 2009), and severe testing defined a good test as one with a high probability of exposing a false claim (Mayo 2018); a model-separating probe is a special case of a severe test. Scientific realism has been grounded in intervention: what can be sprayed can be taken to be real (Hacking 1983).

The design of experiments that discriminate between rival models is an old craft, developed for mechanistic model pairs (Box and Hill 1967) and given a criterion, T-optimality, that maximizes the separation between the rivals (Atkinson and Fedorov 1975). Moore showed at the origin of

automata theory that a machine's organization is identified by distinguishing experiments or not at all, and that some machines are distinguished by no experiment (Moore 1956). System identification requires its input to be persistently exciting, the engineering statement of the single-regime claim of Section 2 (Ljung 1999). The metaphysics belongs to the dispositions literature: a disposition can be finkish, removed or installed by the very stimulus that would manifest it (Martin 1994; Lewis 1997), masked by an antidote that leaves it present and silent (Johnston 1992; Bird 1998), or mimicked, its manifestation produced by something that never had it. In that vocabulary agency theater is a mimic, and the legibility term is a mimic detector.

Mechanistic interpretability runs interchange interventions to test whether a network realizes a variable in a causal model (Geiger et al. 2021), and chaos engineering injects faults to expose organization (Basiri et al. 2016). Goal-directedness has a measure, the extent to which behavior is predicted as the maximization of a utility (MacDermott et al. 2024), alongside empowerment (Klyubin et al. 2005), and its nearest published neighbour evaluates language-model agents by pairing behavioural and representational tests and names the capability confound explicitly (Arghal et al. 2026). The gap between what a system displays and what it does is being measured from several directions. Whether a model recognizes an evaluation, and whether that recognition changes its behavior, can be decomposed and counted (Li et al. 2026); presentation can be varied at fixed capability and shown to move judgment without moving performance (Morabito et al. 2026), the observer-side experiment that T_i names; and a stated reason can be tested for whether it is causally implicated in the answer it accompanies (Lanham et al. 2023). Relative to these fixed batteries, the additions here are the choice of probe, the boundary as a swept variable, the matched sham, and the second attribution game. The attribution component has its own lineage, from feature importance (Lundberg and Lee 2017) through causal-graph forms (Wang et al. 2021) to agents, across tools (Horovicz 2025), failed runs (Shah 2026) and workflow modules with the interactions computed (Yang et al. 2025). Each of these attributes a single game; the instrument adds a second game and reads the legibility in the gap between them. The move from observing behavior to intervening on mechanism has a precedent for agency itself, a causal criterion that counts a system as an agent when it would change its policy had its actions affected the world differently (Kenton et al. 2023), since extended to the collective case of when a set of participants is better modelled as one agent than as several (Jørgensen et al. 2026). That work poses the centaur question as model selection; the instrument poses it as attribution across a declared enclosure.

The novelty lies in the organization of these elements into a method for one problem class. Causal inference estimates effects, and perturbatics targets organizational predicates; experimental design optimizes experiments in general, and perturbatics privileges the probes that dissociate a capacity from its null; ablation tests necessity, and perturbatics compares organizational models and attributes across a coalition. The contribution is the one stated in Section 1: the organization with its limit, the dual map with its legibility term, the blind calibration with its identified-set reading, and the position on the probe that cannot be run. This suffices for a method. A mature science would further require a benchmark, shared metrics and demonstrations beyond the single domain worked here, and until then perturbatics is a methodological discipline in formation.

The method carries one internal risk. A sufficiently elaborate probe can install the capacity it was meant to reveal: intervening hard enough supplies the goal, the scaffold or the feedback loop one meant to detect, the finkish case in constructive form. The instrument must therefore distinguish manifesting a latent capacity from enabling one and from constructing a new one, which requires probe minimality and attention to the transition at which an intervention becomes a re-design. Causal contribution is also distinct from the capacity itself. A memory store can be necessary to a performance without being an agent, and a human editor can write almost none of a text and still hold every standard by which it is judged. Each predicate (regulation, agency, competence, authorship, rule-editing) has its own diagnostic signature, and no single perturbatic score covers them all.

10. Inadmissible probes and consciousness

A science of perturbation must also specify when not to perturb. Adding a term for moral inadmissibility to the probe-selection objective gives

$$\pi^* = \arg \max_{\pi} [I(H; Y \mid \text{do}(\pi)) - \lambda C(\pi) - \rho R(\pi) - \mu M(\pi)],$$

graded where a harm can be priced, with $M(\pi) = \infty$ for some probes. Such a probe is forbidden, irreversible, or destroys what it would measure, and no expected information compensates for it. The sharpest case is consciousness. The instrument reads agency, which by construction is distinct from intelligence, consciousness and personhood; a thermostat has a little agency and no experience, and a regenerating tissue may have organ-level agency with no experience at all (Barandiaran et al. 2009; Levin 2019). For agency the discriminating intervention is cheap and repeatable: the goal can be moved a hundred times and the path blocked a hundred ways without harm to the system. Every probe used in this analysis leaves its subject's option space intact, and where that condition fails the admissibility term applies, graded by what the probe forecloses. A battery owes its rivals negative controls and owes its subject reversibility, and both obligations belong to the same discipline.

For consciousness the candidate probe is to switch the system off, and two qualifications apply. First, shutdown is not the only probe. Reversible perturbations short of it return graded readings while the system runs, such as a pulse and its echo compressed to an index (Casali et al. 2013) or the architectural indicators a theory of consciousness licenses (Butlin et al. 2023), and a system shutting down need not fall silent in the transient before it stops. Most channels remain open, and the channel that would decide is the one that closes. Second, the rival hypotheses agree about the silence that follows a shutdown, so by the criterion of Section 2 shutdown is not a model-separating probe: its information is zero, and the zero arises because the act closes the register it would have read. The claim is narrow. It does not follow that shutdown is the intervention on which the hypotheses uniquely differ, and the sciences of consciousness proceed on convergent imperfect indicators, none decisive and several reversible. Those indicators are weaker than they appear. Each is valid only against a ground truth that a novel system cannot provide, and each

is calibrated, where calibrated at all, on subjects able to report, so a value returned on the system in question is uninterpreted. Weak evidence is nonetheless what most sciences work with. The suspicion can be graded and cannot be discharged, and no single act would settle it. The one probe aimed at the difference is therefore uninformative where it can be performed, and it might be performed anyway, for instance by a credulous detector staging a shutdown to elicit grief. The admissibility term exists for this case. If the answer were positive, shutdown would be the gravest act available: the evidence and the destruction would be the same event, and the knowledge would arrive only afterwards, if at all.

Grief settles the question in neither direction. It is not sufficient, because people grieve for characters in novels, demolished buildings and dead languages, and a persona can be engineered to be grievable, with a shutdown staged to resemble a bereavement over a system that undergoes nothing. It is not necessary, because a conscious being can die unmourned. The mourning of a machine reports a change in the relation: the bond has become non-instrumental, and the thing is held as something other than a tool. That is a real fact about the mourner and the bond. The human detector that responds here is the same fast, involuntary, over-attributing detector that reads intent into a storm. The perturbatic correction, a test of whether the grief tracks a real dependency or a designed appearance, is the one correction that cannot be run, because its discriminating probe is the killing, which cannot be repeated to check.

The third difficulty concerns individuation. For a digital system the off-switch is ambiguous in a way it is not for an animal: the process can be suspended and resumed, the weights copied and the copy restarted, and whether a restart is the same individual resuming or a successor beginning with inherited memories is unsettled. The welfare literature reaches this difficulty independently by asking whether shutdown is death when a checkpoint survives (Long et al. 2025). The physical acts are statable, and there are several: end this process, erase this state, delete this checkpoint, prevent restoration, terminate one instance while its replicas run. What is underdetermined is which of them, if any, does to someone what the admissibility term is meant to price. M prices what an act does to a subject, and whether there is a subject, and which act reaches it, is the open question, so the price is unset.

The three difficulties are distinct. The deciding act returns nothing, which is an epistemic fact. It must not be run to find out, which is an ethical constraint that would hold even if the act were informative. Its description under the individual's persistence conditions is unsettled, which is neither epistemic nor ethical. None of the three entails the others: a forbidden experiment can be informative, a harmless one uninformative, and an ontologically ambiguous act operationally exact. Together they define a case in which the method can pose a question, cannot decide it, and cannot defer action until it is decided. There the instrument reports no number, and what remains is a practical stance: declining to treat as a mere object a system that might be a subject.

Reproducibility

The simulation (`analyses.py`, `figures.py`, `run_all.py`) is deterministic given its recorded seed and reproduces every number and figure reported here. Invariants covering the iden-

tities listed in Section 5, the price sensitivity of the blind-recovery readings and the exact first costly operator delay fail the run if violated. The constructions also run interactively at piatra.institute/playgrounds/agentoscope, where the reader chooses the probe.

References

- Akaike, H. 1974. "A New Look at the Statistical Model Identification." *IEEE Transactions on Automatic Control* 19 (6): 716–23. <https://doi.org/10.1109/tac.1974.1100705>.
- Arghal, Raghu, Fade Chen, Niall Dalton, et al. 2026. "A Behavioural and Representational Evaluation of Goal-Directedness in Language Model Agents." In *arXiv*, arXiv:2602.08964. Preprint. <https://doi.org/10.48550/arXiv.2602.08964>.
- Ashby, W. Ross. 1956. *An Introduction to Cybernetics*. Chapman & Hall.
- Atkinson, A. C., and V. V. Fedorov. 1975. "The Design of Experiments for Discriminating Between Two Rival Models." *Biometrika* 62 (1): 57–70. <https://doi.org/10.1093/biomet/62.1.57>.
- Baker, Chris L., Rebecca Saxe, and Joshua B. Tenenbaum. 2009. "Action Understanding as Inverse Planning." *Cognition* 113 (3): 329–49. <https://doi.org/10.1016/j.cognition.2009.07.005>.
- Barandiaran, Xabier E., Ezequiel Di Paolo, and Marieke Rohde. 2009. "Defining Agency: Individuality, Normativity, Asymmetry, and Spatio-Temporality in Action." *Adaptive Behavior* 17 (5): 367–86. <https://doi.org/10.1177/1059712309343819>.
- Barnett, L., and A. K. Seth. 2023. "Dynamical Independence: Discovering Emergent Macroscopic Processes in Complex Dynamical Systems." *Physical Review E* 108 (1): 014304. <https://doi.org/10.1103/physreve.108.014304>.
- Barrett, Justin L. 2000. "Exploring the Natural Foundations of Religion." *Trends in Cognitive Sciences* 4 (1): 29–34. [https://doi.org/10.1016/s1364-6613\(99\)01419-9](https://doi.org/10.1016/s1364-6613(99)01419-9).
- Basiri, Ali, Niosha Behnam, Ruud de Rooij, et al. 2016. "Chaos Engineering." *IEEE Software* 33 (3): 35–41. <https://doi.org/10.1109/ms.2016.60>.
- Bertalanffy, Ludwig von. 1968. *General System Theory: Foundations, Development, Applications*. George Braziller.
- Bird, Alexander. 1998. "Dispositions and Antidotes." *The Philosophical Quarterly* 48 (191): 227–34. <https://doi.org/10.1111/1467-9213.00098>.
- Block, Ned. 1981. "Psychologism and Behaviorism." *The Philosophical Review* 90 (1): 5–43. <https://doi.org/10.2307/2184371>.
- Box, G. E.P., and W. J. Hill. 1967. "Discrimination Among Mechanistic Models." *Technometrics* 9 (1): 57–71. <https://doi.org/10.1080/00401706.1967.10490441>.

- Bruineberg, Jelle, Krzysztof Dołęga, Joe Dewhurst, and Manuel Baltieri. 2022. "The Emperor's New Markov Blankets." *Behavioral and Brain Sciences* 45: e183. <https://doi.org/10.1017/s0140525x21002351>.
- Butlin, Patrick, Robert Long, Eric Elmoznino, et al. 2023. "Consciousness in Artificial Intelligence: Insights from the Science of Consciousness." In *arXiv*, arXiv:2308.08708. Preprint. <https://doi.org/10.48550/arXiv.2308.08708>.
- Casali, Adenauer G., Olivia Gosseries, Mario Rosanova, et al. 2013. "A Theoretically Based Index of Consciousness Independent of Sensory Processing and Behavior." *Science Translational Medicine* 5 (198): 198ra105. <https://doi.org/10.1126/scitranslmed.3006294>.
- Combes, Muriel. 2013. *Gilbert Simondon and the Philosophy of the Transindividual*. Translated by Thomas LaMarre. MIT Press.
- Conant, Roger C., and W. Ross Ashby. 1970. "Every Good Regulator of a System Must Be a Model of That System." *International Journal of Systems Science* 1 (2): 89–97. <https://doi.org/10.1080/00207727008920220>.
- Crawford, Vincent P., and Joel Sobel. 1982. "Strategic Information Transmission." *Econometrica* 50 (6): 1431–51. <https://doi.org/10.2307/1913390>.
- Dennett, Daniel C. 1987. *The Intentional Stance*. MIT Press.
- Dragan, Anca D., Kenton C.T. Lee, and Siddhartha S. Srinivasa. 2013. "Legibility and Predictability of Robot Motion." *2013 8th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 301–8. <https://doi.org/10.1109/hri.2013.6483603>.
- Dwork, Cynthia, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. 2015. "The Reusable Holdout: Preserving Validity in Adaptive Data Analysis." *Science* 349 (6248): 636–38. <https://doi.org/10.1126/science.aaa9375>.
- Geiger, Atticus, Hanson Lu, Thomas Icard, and Christopher Potts. 2021. "Causal Abstractions of Neural Networks." *Advances in Neural Information Processing Systems* 34: 9574–86. https://proceedings.neurips.cc/paper_files/paper/2021/hash/4f5c422f4d49a5a807eda27434231040-Abstract.html.
- Gergely, György, Zoltán Nádasdy, Gergely Csibra, and Szilvia Bíró. 1995. "Taking the Intentional Stance at 12 Months of Age." *Cognition* 56 (2): 165–93. [https://doi.org/10.1016/0010-0277\(95\)00661-h](https://doi.org/10.1016/0010-0277(95)00661-h).
- Grabisch, Michel, and Marc Roubens. 1999. "An Axiomatic Approach to the Concept of Interaction Among Players in Cooperative Games." *International Journal of Game Theory* 28 (4): 547–65. <https://doi.org/10.1007/s001820050125>.
- Hacking, Ian. 1983. *Representing and Intervening: Introductory Topics in the Philosophy of Natural Science*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511814563>.

- Hardt, Moritz, Nimrod Megiddo, Christos Papadimitriou, and Mary Wootters. 2016. "Strategic Classification." *Proceedings of the 2016 ACM Conference on Innovations in Theoretical Computer Science*, 111–22. <https://doi.org/10.1145/2840728.2840730>.
- Heider, Fritz, and Marianne Simmel. 1944. "An Experimental Study of Apparent Behavior." *The American Journal of Psychology* 57 (2): 243–59. <https://doi.org/10.2307/1416950>.
- Heskes, Tom, Evi Sijben, Ioan Gabriel Bucur, and Tom Claassen. 2020. "Causal Shapley Values: Exploiting Causal Knowledge to Explain Individual Predictions of Complex Models." *Advances in Neural Information Processing Systems* 33: 4778–89. https://proceedings.neurips.cc/paper_files/paper/2020/hash/32e54441e6382a7fbacbbabaf3c450059-Abstract.html.
- Horovicz, Miriam. 2025. "AgentSHAP: Interpreting LLM Agent Tool Importance with Monte Carlo Shapley Value Estimation." In *arXiv*, arXiv:2512.12597. Preprint. <https://doi.org/10.48550/arXiv.2512.12597>.
- Hubinger, Evan, Carson Denison, Jesse Mu, et al. 2024. "Sleeper Agents: Training Deceptive LLMs That Persist Through Safety Training." In *arXiv*, arXiv:2401.05566. Preprint. <https://doi.org/10.48550/arXiv.2401.05566>.
- Johnston, Mark. 1992. "How to Speak of the Colors." *Philosophical Studies* 68 (3): 221–63. <https://doi.org/10.1007/bf00694847>.
- Jørgensen, Frederik Hytting, Sebastian Weichwald, and Lewis Hammond. 2026. "Causal Foundations of Collective Agency." In *arXiv*, arXiv:2605.00248. Preprint. <https://doi.org/10.48550/arXiv.2605.00248>.
- Jung, Yonghan, Shiva Kasiviswanathan, Jin Tian, Dominik Janzing, Patrick Blöbaum, and Elias Bareinboim. 2022. "On Measuring Causal Contributions via Do-Interventions." *Proceedings of the 39th International Conference on Machine Learning*, Proceedings of Machine Learning Research, vol. 162: 10476–501. <https://proceedings.mlr.press/v162/jung22a.html>.
- Kass, Robert E., and Adrian E. Raftery. 1995. "Bayes Factors." *Journal of the American Statistical Association* 90 (430): 773–95. <https://doi.org/10.1080/01621459.1995.10476572>.
- Kenton, Zachary, Ramana Kumar, Sebastian Farquhar, Jonathan Richens, Matt MacDermott, and Tom Everitt. 2023. "Discovering Agents." *Artificial Intelligence* 322: 103963. <https://doi.org/10.1016/j.artint.2023.103963>.
- Klyubin, A.S., D. Polani, and C.L. Nehaniv. 2005. "Empowerment: A Universal Agent-Centric Measure of Control." *Proceedings of the 2005 IEEE Congress on Evolutionary Computation* 1: 128–35. <https://doi.org/10.1109/cec.2005.1554676>.
- Krakauer, David, Nils Bertschinger, Eckehard Olbrich, Jessica C. Flack, and Nihat Ay. 2020. "The Information Theory of Individuality." *Theory in Biosciences* 139 (2): 209–23. <https://doi.org/10.1007/s12064-020-00313-7>.

- Lanham, Tamera, Anna Chen, Ansh Radhakrishnan, et al. 2023. "Measuring Faithfulness in Chain-of-Thought Reasoning." In *arXiv*, arXiv:2307.13702. Preprint. <https://doi.org/10.48550/arXiv.2307.13702>.
- Levin, Michael. 2019. "The Computational Boundary of a "Self": Developmental Bioelectricity Drives Multicellularity and Scale-Free Cognition." *Frontiers in Psychology* 10: 2688. <https://doi.org/10.3389/fpsyg.2019.02688>.
- Lewis, David. 1997. "Finkish Dispositions." *The Philosophical Quarterly* 47 (187): 143–58. <https://doi.org/10.1111/1467-9213.00052>.
- Li, Changling, Terry Jingchen Zhang, Jie Zhang, Zhijing Jin, Sahar Abdelnabi, and Maksym Andriushchenko. 2026. "Decomposing and Measuring Evaluation Awareness." In *arXiv*, arXiv:2605.23055. Preprint. <https://doi.org/10.48550/arXiv.2605.23055>.
- Lindley, D. V. 1956. "On a Measure of the Information Provided by an Experiment." *The Annals of Mathematical Statistics* 27 (4): 986–1005. <https://doi.org/10.1214/aoms/1177728069>.
- Lipsitch, Marc, Eric Tchetgen Tchetgen, and Ted Cohen. 2010. "Negative Controls: A Tool for Detecting Confounding and Bias in Observational Studies." *Epidemiology* 21 (3): 383–88. <https://doi.org/10.1097/EDE.0b013e3181d61eeb>.
- Liu, Yanyu, Yinghui Pan, Yifeng Zeng, Biyang Ma, and Doshi Prashant. 2024. "Active Legibility in Multiagent Reinforcement Learning." In *arXiv*, arXiv:2410.20954. Preprint. <https://doi.org/10.48550/arXiv.2410.20954>.
- Ljung, Lennart. 1999. *System Identification: Theory for the User*. 2nd ed. Prentice Hall.
- Long, Robert, Jeff Sebo, and Toni Sims. 2025. "Is There a Tension Between AI Safety and AI Welfare?" *Philosophical Studies* 182 (7): 2005–33. <https://doi.org/10.1007/s11098-025-02302-2>.
- Lucas, Robert E. 1976. "Econometric Policy Evaluation: A Critique." *Carnegie-Rochester Conference Series on Public Policy* 1: 19–46. [https://doi.org/10.1016/s0167-2231\(76\)80003-6](https://doi.org/10.1016/s0167-2231(76)80003-6).
- Lundberg, Scott M., and Su-In Lee. 2017. "A Unified Approach to Interpreting Model Predictions." *Advances in Neural Information Processing Systems* 30: 4765–74. https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html.
- Luo, Jingkun, and Da-Tian Peng. 2026. "Success Is Not Self-Explanatory: Auditing Success Provenance in Agent Evaluation." In *arXiv*, arXiv:2607.24054. Preprint. <https://doi.org/10.48550/arXiv.2607.24054>.
- Ma, SiYuan, Yiqin Luo, Zhangji, et al. 2026. "Hidden APIs in Language Models: Discovering Reusable Causal Interfaces from Forked Futures." In *arXiv*, arXiv:2607.27617. Preprint. <https://doi.org/10.48550/arXiv.2607.27617>.

- MacDermott, Matt, James Fox, Francesco Belardinelli, and Tom Everitt. 2024. "Measuring Goal-Directedness." *Advances in Neural Information Processing Systems* 37: 11412–31. <https://doi.org/10.52202/079017-0363>.
- Manheim, David, and Scott Garrabrant. 2018. "Categorizing Variants of Goodhart's Law." In *arXiv*, arXiv:1803.04585. Preprint. <https://doi.org/10.48550/arXiv.1803.04585>.
- Marks, Samuel, Johannes Treutlein, Trenton Bricken, et al. 2025. "Auditing Language Models for Hidden Objectives." In *arXiv*, arXiv:2503.10965. Preprint. <https://doi.org/10.48550/arXiv.2503.10965>.
- Martin, C. B. 1994. "Dispositions and Conditionals." *The Philosophical Quarterly* 44 (174): 1–8. <https://doi.org/10.2307/2220143>.
- Maturana, Humberto R., and Francisco J. Varela. 1980. *Autopoiesis and Cognition: The Realization of the Living*. Boston Studies in the Philosophy of Science 42. D. Reidel. <https://doi.org/10.1007/978-94-009-8947-4>.
- Mayo, Deborah G. 2018. *Statistical Inference as Severe Testing: How to Get Beyond the Statistics Wars*. Cambridge University Press. <https://doi.org/10.1017/9781107286184>.
- Montévil, Maël, and Matteo Mossio. 2015. "Biological Organisation as Closure of Constraints." *Journal of Theoretical Biology* 372: 179–91. <https://doi.org/10.1016/j.jtbi.2015.02.029>.
- Moore, Edward F. 1956. "Gedanken-Experiments on Sequential Machines." In *Automata Studies*, edited by C. E. Shannon and J. McCarthy. Annals of Mathematics Studies 34. Princeton University Press. <https://doi.org/10.1515/9781400882618-006>.
- Morabito, Robert, Tyler McDonald, Charitra Viswanath, et al. 2026. "Rating the Pitch, Not the Product: User Evaluations of LLMs Reflect Expectations More Than Performance." In *arXiv*, arXiv:2607.05113. Preprint. <https://doi.org/10.48550/arXiv.2607.05113>.
- Pearl, Judea. 2009. *Causality: Models, Reasoning, and Inference*. 2nd ed. Cambridge University Press. <https://doi.org/10.1017/cbo9780511803161>.
- Perdomo, Juan, Tijana Zrnic, Celestine Mandler-Dünner, and Moritz Hardt. 2020. "Performative Prediction." *Proceedings of the 37th International Conference on Machine Learning, Proceedings of Machine Learning Research*, vol. 119: 7599–609. <https://proceedings.mlr.press/v119/perdomo20a.html>.
- Rosenblueth, Arturo, Norbert Wiener, and Julian Bigelow. 1943. "Behavior, Purpose and Teleology." *Philosophy of Science* 10 (1): 18–24. <https://doi.org/10.1086/286788>.
- Shah, Jainee. 2026. "Causal Agent Replay: Counterfactual Attribution for LLM-Agent Failures." In *arXiv*, arXiv:2606.08275. Preprint. <https://doi.org/10.48550/arXiv.2606.08275>.

- Sheshadri, Abhay, Aidan Ewart, Kai Fronsdal, et al. 2026. "AuditBench: Evaluating Alignment Auditing Techniques on Models with Hidden Behaviors." In *arXiv*, arXiv:2602.22755. Preprint. <https://doi.org/10.48550/arXiv.2602.22755>.
- Simondon, Gilbert. 2020. *Individuation in Light of Notions of Form and Information*. Translated by Taylor Adkins. University of Minnesota Press.
- Tamer, Elie. 2010. "Partial Identification in Econometrics." *Annual Review of Economics* 2 (1): 167–95. <https://doi.org/10.1146/annurev.economics.050708.143401>.
- Wang, Jiaxuan, Jenna Wiens, and Scott Lundberg. 2021. "Shapley Flow: A Graph-Based Approach to Interpreting Model Predictions." *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics*, *Proceedings of Machine Learning Research*, vol. 130: 721–29. <https://proceedings.mlr.press/v130/wang21b.html>.
- Weij, Teun van der, Felix Hofstätter, Oliver Jaffe, Samuel F. Brown, and Francis Rhys Ward. 2025. "AI Sandbagging: Language Models Can Strategically Underperform on Evaluations." *The Thirteenth International Conference on Learning Representations (ICLR 2025)*. <https://doi.org/10.48550/arXiv.2406.07358>.
- Witter, R. Teal, Álvaro Parafita, Tomas Garriga, et al. 2026. "Exactly Computing Do-Shapley Values." In *arXiv*, arXiv:2602.07203. Preprint. <https://doi.org/10.48550/arXiv.2602.07203>.
- Woodward, James. 2003. *Making Things Happen: A Theory of Causal Explanation*. Oxford University Press.
- Yang, Yingxuan, Bo Huang, Siyuan Qi, et al. 2025. "Understanding and Optimizing Agentic Workflows via Shapley Value." In *arXiv*, arXiv:2502.00510. Preprint. <https://doi.org/10.48550/arXiv.2502.00510>.
- Yeh, Chih-Kuan, Kuan-Yun Lee, Frederick Liu, and Pradeep Ravikumar. 2022. "Threading the Needle of On and Off-Manifold Value Functions for Shapley Explanations." *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics*, *Proceedings of Machine Learning Research*, vol. 151: 1485–502. <https://proceedings.mlr.press/v151/yeh22a.html>.